



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION

AMRITA
VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

**Virtual
Labs**
An MoE Govt of India Initiative



 **VALUE VIRTUAL LABS**
broadening perspectives

AMRITA
VIRTUAL LABS
Broadening Perspectives

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OUR INSPIRATION

As a globally revered spiritual and humanitarian leader and the Chancellor of Amrita Vishwa Vidyapeetham, Mata Amritanandamayi (Amma) guides the University with the profound vision of Vasudhaiva Kutumbakam—the understanding that the world is one family. She reminds us that true education must extend beyond physical comfort and intellectual competition to awaken the compassion, discernment, and latent potential hidden within the Inner Self. Her message is an urgent invitation to transform our lives into an expression of selfless service, urging us to become the compassionate hands of the divine in a changing world



Sri Mata Amritanandamayi Devi (AMMA)
Chancellor, Amrita Vishwa Vidyapeetham

ABOUT AMRITA VIRTUAL LABS

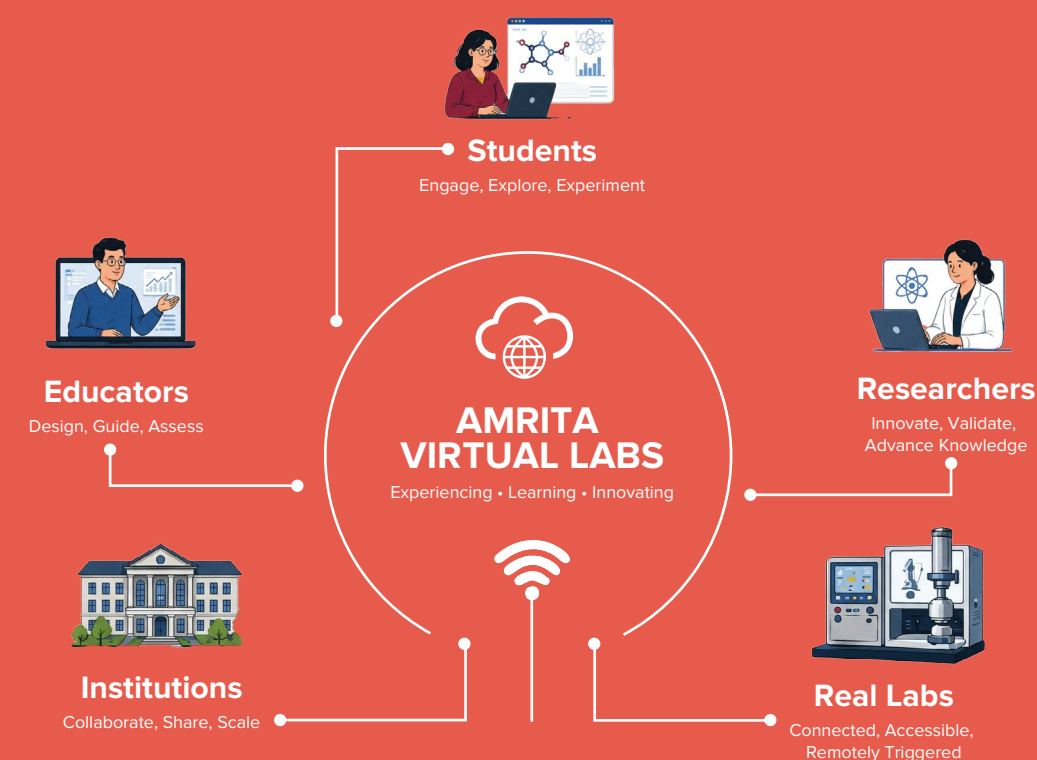
Amrita Virtual Labs is one of the world's largest collections of interactive virtual laboratories, developed under the National Mission on Education through ICT (NMEICT), Ministry of Education, Government of India.

The initiative democratizes access to high-quality laboratory education by enabling students to perform experiments anytime, anywhere through immersive simulations, animations, and remotely triggered laboratories.

REIMAGINING LABORATORY EDUCATION

FOR THE DIGITAL ERA

Universalizing access to high-quality experimental learning through technology, research, and innovation.



CLOUD PLATFORM

Always Available
Access high-quality virtual labs anytime, anywhere.



AI ASSISTED

Learning Support
Intelligent guidance and adaptive support for deeper learning.



SECURE & RELIABLE

Global Standards
Enterprise-grade security and reliability you can trust.



OPEN & COLLABORATIVE

Knowledge for All
Collaborating across institutions to create and share knowledge.



DATA & ANALYTICS

Insights for Improvement
Data-driven insights to enhance teaching, learning, and outcomes.

WHY LABORATORY EDUCATION NEEDS TRANSFORMATION

Practical laboratory education is the foundation of science, engineering, medicine, and technology. It enables learners to develop experimental skills, critical thinking, problem-solving abilities, and scientific reasoning that cannot be acquired through classroom instruction alone.

However, delivering high-quality laboratory education at scale remains one of the greatest challenges faced by educational institutions across the world.



GLOBAL LEARNING
Learning Without Boundaries

Amrita Virtual Labs addresses these challenges by providing immersive, interactive, and research-driven laboratory experiences through simulations, animations, and remotely triggered experiments.

Rather than replacing conventional laboratories, Virtual Labs extend and enrich laboratory education by making high-quality experimental learning accessible, scalable, inclusive, and available anytime, anywhere.



“Virtual Laboratories bridge the gap between theory and practice by making high-quality experimental learning accessible, scalable, and research-driven for every learner, everywhere.”

THE TRANSFORMATION JOURNEY



KEY CHALLENGES

INFRASTRUCTURE & RESOURCE SCARCITY

Over 86% of institutions report a shortage of apparatus and high equipment costs as primary educational hurdles

OPERATIONAL & SAFETY CONSTRAINTS

Large student batches and safety concerns often limit hands-on access to advanced scientific instruments.

GEOGRAPHICAL & TIME LIMITATIONS

Fixed laboratory hours and physical location requirements create significant barriers for remote and time-constrained learners.

DEFINITION: THE CONNECTED LEARNING ECOSYSTEM

A cloud-based platform integrating AI-assisted learning, simulations, and remote-accessible hardware for global collaboration.

AMRITA SOLUTIONS

PROCESS STEP: INQUIRY-DRIVEN INTERACTIVE PEDAGOGY

Enables anytime-anywhere access to interactive simulations that allow students to vary parameters and visualize impossible phenomena.

STATISTIC: PROVEN EDUCATIONAL IMPACT AT SCALE

Serving 11.3M+ users across 140+ countries, significantly increasing student experimental self-efficacy and conceptual understanding.



AMRITA VIRTUAL LABS AT A GLANCE

One of the world's largest digital laboratory ecosystems for science and engineering education.

Designed to complement conventional laboratory instruction, Amrita Virtual Labs enables learners to explore scientific concepts through safe, interactive, and inquiry-driven experimental experiences. By combining

simulations, animations, and remotely triggered experiments, the platform strengthens conceptual understanding, develops practical competencies, and

enhances laboratory preparedness before hands-on experimentation.

Today, Amrita Virtual Labs supports millions of learners and hundreds of educational institutions, advancing experiential learning through innovation, accessibility, and global collaboration. With a rich portfolio of digital learning resources spanning science and engineering disciplines, the platform fosters curiosity, critical thinking, experimentation, and lifelong learning while making high-quality

"More than a virtual laboratory platform, Amrita Virtual Labs is a research-driven educational ecosystem that is redefining experimental learning and expanding access to world-class laboratory education."

laboratory education available beyond the boundaries of time and location.

OUTREACH OF AMRITA VIRTUAL LABS



276

NODAL CENTRES
Across India



2,36,237

REGISTERED USERS
On Amrita Virtual
Labs



44,08,490

UNIQUE VISITORS
From around the
world



2,950

**WORKSHOPS
CONDUCTED**
Across India



1,03,737

STUDENTS TRAINED
Through outreach
programs



69,158

TEACHERS TRAINED
Empowering
educators



145

**ENGINEERING
COLLEGES**
Nodal Centres



131

**ARTS & SCIENCE
COLLEGES**
Nodal Centres



18

STATES COVERED
Pan-India Presence



140+

**COUNTRIES
REACHED**
Global Impact



257+

**TOTAL
DIGITAL ASSETS**



495

**TOTAL
EXPERIMENTS**

THE ECOSYSTEM

Amrita Virtual Labs offers a comprehensive portfolio of digital laboratory resources spanning multiple science and engineering disciplines. Each discipline integrates simulations, animations, interactive visualizations, and instructional videos

222
86.4%

SIMULATIONS

Interactive experimentation
in a virtual environment.

PORTFOLIO HIGHLIGHTS

81

Experiments

PHYSICS

Interactive simulations and visual demonstrations for understanding fundamental physical phenomena.

38

Experiments

CHEMISTRY

Digital experiments and animations that simplify chemical concepts and laboratory procedures.

239

Experiments

BIOTECHNOLOGY

Comprehensive simulation-based learning resources covering biological processes, laboratory techniques, and experimental workflows.

110

Experiments

COMPUTER SCIENCE

Hands-on virtual laboratories for programming, algorithms, networking, cybersecurity, and emerging computing technologies.

to support conceptual understanding, experimental exploration, and practical skill development. Designed using evidence-based instructional approaches, these resources enable learners to engage with complex scientific concepts through immersive, self-paced, and inquiry-driven learning experiences.

SUPPORTING MULTIPLE LEARNING MODALITIES

The discipline portfolio combines diverse digital resources to address different learning needs:

38
14.8%

ANIMATIONS

Visualization of abstract and dynamic scientific concepts.

6
2.3%

INTERACTIVE ANIMATIONS

Guided exploration through learner interaction

59
22.9%

INSTRUCTIONAL VIDEOS

Step-by-step demonstrations and conceptual explanations.



RESEARCH IMPACT & SCIENTIFIC CONTRIBUTIONS

Amrita Virtual Labs has contributed significantly to the advancement of virtual laboratory research through peer-reviewed publications, educational studies, international collaborations, and evidence-based evaluation. Research conducted using the platform has demonstrated improvements in conceptual understanding, learner engagement, laboratory preparedness, experimental self-efficacy, and reflective learning across diverse educational settings.

OUR RESEARCH FOCUS



EDUCATIONAL TECHNOLOGY

Developing innovative approaches for technology-enhanced laboratory learning.



LEARNING SCIENCES

Understanding how learners engage, explore and learn through simulations, experimentation and visualization.

KEY RESEARCH THEMES



Virtual Laboratory Pedagogy



Experimental Self-Efficacy



Reflective Learning



Learning Analytics



Remote Laboratories



STEM Education



Digital Learning Ecosystems

SELECTED PUBLICATIONS



The role of integrating augmented reality and cognitive constructivism in enhancing sustainable education
Educ Inf Technol 30, 26151–26186 (2025).



Virtual labs in chemistry education: A novel approach for increasing student's laboratory educational consciousness and skills
Educ Inf Technol 29, 25307–25331 (2024)

100+ Publications

30 Journal articles

53 Conference papers

9 Book Chapters

11 Poster presentations



Intrinsic and extrinsic motivation among students for laboratory courses - Assessing the impact of virtual laboratories
Computers & Education 198 (2023): 104758.



Virtual experimental platforms in chemistry laboratory education and its impact on experimental self-efficacy
Int J Educ Technol High Educ 17, 30 (2020).



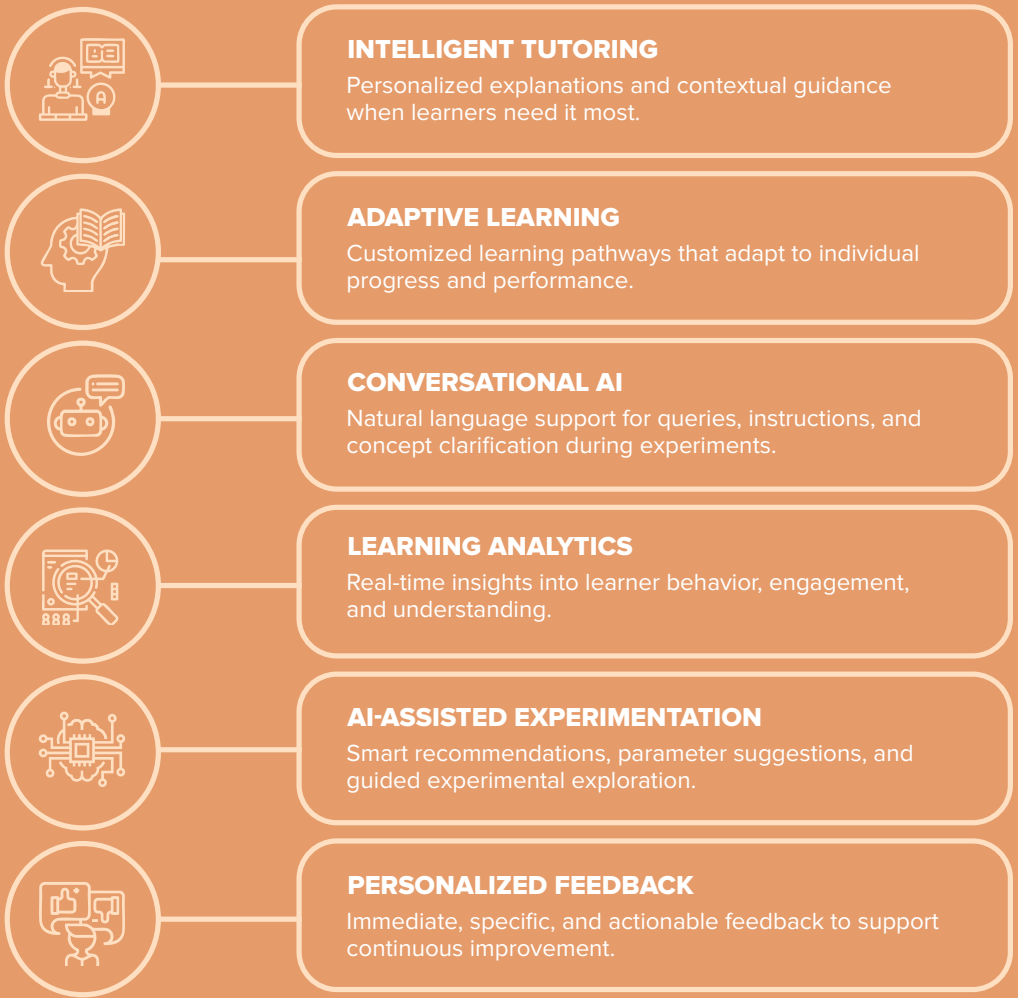
Impact of remote experimentation, interactivity and platform effectiveness on laboratory learning outcomes
Int J Educ Technol High Educ 18, 38 (2021).

“Research is not an outcome of Amrita Virtual Labs—it is the foundation upon which every innovation is designed, evaluated, and continuously improved.”

ARTIFICIAL INTELLIGENCE FOR THE NEXT GENERATION OF VIRTUAL LABORATORIES

Artificial Intelligence is redefining digital laboratory education by enabling learning experiences that are adaptive, personalized, and data-driven. Building upon years of educational research, Amrita Virtual Labs is integrating AI technologies to provide intelligent guidance, personalized feedback, adaptive experimentation, and learning analytics that support every learner throughout the laboratory learning journey. This evolution transforms Virtual Labs from digital experiment repositories into intelligent learning companions that foster deeper understanding, learner autonomy and lifelong learning.

AI CAPABILITIES IN VIRTUAL LABS



ARTIFICIAL INTELLIGENCE

COGNITIVE PSYCHOLOGICAL & BEHAVIORAL FACTORS

STUDIED IN RESEARCH PUBLICATIONS

Understanding how learners think, feel and act in virtual laboratory learning environments



NATIONAL MISSION ON EDUCATION THROUGH ICT

Formation of Consortia - Development and realization of Virtual Laboratories



IIT Delhi



IIT Roorkee



AMRITA
VISHWA VIDYAPEETHAM



IIT Guwahati



IIT Bombay



IIIT Hyderabad



IIT Kanpur



Dayalbag Educational Institute



IIT Kharagpur



NIT Karnataka



COE Pune



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